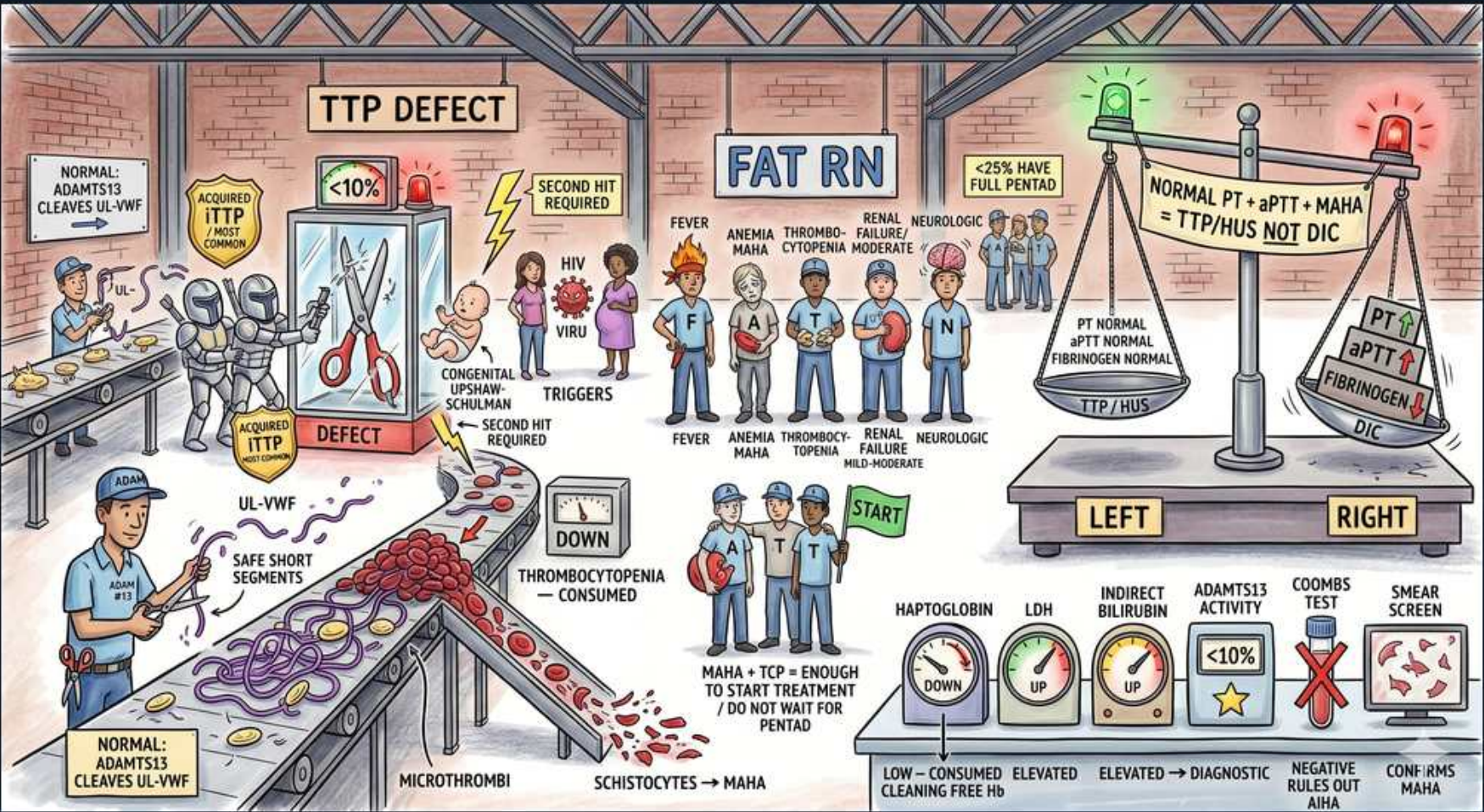


Platelet disorders — TTP (TTP Defect factory)



1. ADAMTS13 deficiency

WHAT YOU SEE

Scissors at a 'TTP DEFECT' gate labeled 'NORMAL: ADAMTS13 CLEAVES UL-vWF' but stamped '<10%' — the cutting scissors are the ADAMTS13 enzyme, here broken so the giant vWF strings go uncut.

WHAT IT ENCODES

TTP is caused by severely deficient ADAMTS13 activity (<10%). ADAMTS13 is the metalloprotease that normally cleaves ultra-large von Willebrand factor (UL-vWF) multimers as they are released from endothelium. Without it, UL-vWF strings persist, bind and clump platelets in high-shear microvasculature, and form platelet-rich microthrombi — driving consumption and mechanical RBC shearing.

BOARD TRAP

A microangiopathy with NORMAL ADAMTS13 activity (>10%) is NOT TTP — think HUS (Shiga-toxin or complement-mediated), DIC, malignant hypertension, or drug-induced TMA instead. The discriminator the boards love: ADAMTS13 <10% defines TTP; preserved ADAMTS13 with a diarrheal prodrome and prominent renal failure in a child points to HUS.

2. Acquired (autoantibody)

WHAT YOU SEE

The 'ACQUIRED ITP / DEFECT' gate next to the congenital Upshaw-Schulman one — the acquired arm is the antibody that disables the ADAMTS13 scissors, the most common adult cause.

WHAT IT ENCODES

Most TTP (especially in adults) is ACQUIRED/immune: an IgG AUTOANTIBODY (inhibitor) against ADAMTS13 neutralizes the enzyme. This is why immunosuppression (steroids, rituximab) is part of treatment in addition to plasma exchange. It peaks in adults, with a female predominance.

BOARD TRAP

Assuming all TTP is inherited is the trap — the congenital form (Upshaw-Schulman, ADAMTS13 gene mutation) is rare and presents in neonates/children or in pregnancy. In a previously healthy ADULT presenting acutely, the answer is ACQUIRED autoantibody-mediated TTP, treated with PLEX plus immunosuppression — not gene therapy or plasma infusion alone.

3. Triggers / second hit

WHAT YOU SEE

Lightning bolts marked 'SECOND HIT REQUIRED' with HIV, pregnancy, and drug figures — the trigger sparks that unmask TTP.

WHAT IT ENCODES

Acute TTP often needs a 'second hit' to unmask it: common triggers include HIV (always test), pregnancy/postpartum, certain drugs, infection, and autoimmune disease. The congenital form (Upshaw-Schulman) is similarly precipitated by pregnancy or illness because baseline ADAMTS13 is already low.

BOARD TRAP

Missing HIV as a precipitant/association of TTP is a classic omission — every new TTP gets an HIV test. The trap stem describes a young patient with TTP and risk factors but you forget to test HIV; in pregnancy, the mirror trap is reflexively calling microangiopathy 'TTP' when HELLP or preeclampsia may better fit (check LFTs, BP, ADAMTS13).

4. Fever

WHAT YOU SEE

The first 'FAT RN' worker on the line holding a thermometer — the F = Fever, the least reliable pentad member.

WHAT IT ENCODES

Fever is the 'F' of the classic pentad (FAT RN = Fever, Anemia [MAHA], Thrombocytopenia, Renal, Neurologic). It is the LEAST consistent feature and is frequently absent. Its presence supports TTP but its absence absolutely does not exclude it.

BOARD TRAP

Requiring fever (or any specific pentad member) before diagnosing/treating TTP is the trap. Fever is often missing; demanding it before starting plasma exchange delays life-saving therapy. The threshold to act is MAHA + thrombocytopenia without another explanation — not the complete FAT RN.

5. MAHA anemia

WHAT YOU SEE

The 'FAT RN' worker labeled ANEMIA / MAHA standing over fragmented red cells — the A = microangiopathic hemolytic anemia from sheared RBCs.

WHAT IT ENCODES

The 'A' = microangiopathic hemolytic anemia: RBCs are physically sheared as they squeeze past platelet-vWF microthrombi, producing SCHISTOCYTES on the smear plus hemolysis labs (high LDH, low haptoglobin, high indirect bilirubin, elevated reticulocytes) and a NEGATIVE Coombs.

BOARD TRAP

Calling a Coombs-POSITIVE hemolytic anemia 'TTP' is wrong — TTP hemolysis is mechanical and therefore Coombs/DAT NEGATIVE. A positive Coombs points to autoimmune hemolysis (warm AIHA, or Evans syndrome when paired with ITP), a completely different management pathway (steroids, not PLEX).

6. Thrombocytopenia

WHAT YOU SEE

The 'FAT RN' worker labeled THROMBOCYTOPENIA beside a DOWN arrow on the platelet counter — the T = consumed platelets, with normal coags.

WHAT IT ENCODES

The 'T' = consumptive thrombocytopenia: platelets are used up forming the diffuse microthrombi, so counts fall (often profoundly, frequently <30,000). Importantly, this consumption occurs with NORMAL coagulation times — unlike DIC, the clotting cascade is not driving it.

BOARD TRAP

Interpreting TTP thrombocytopenia as a clotting-cascade consumptive coagulopathy (DIC) is the trap. In TTP the PT/aPTT/fibrinogen are NORMAL; if those are deranged with low fibrinogen and high D-dimer, the answer is DIC, not TTP. The platelet drop in TTP is from microthrombotic consumption with an intact cascade.

7. Renal + neuro

WHAT YOU SEE

The 'FAT RN' workers labeled RENAL (FAILURE MODERATE) and NEUROLOGIC — the R and N, with neuro dominant and renal only mild-moderate in TTP.

WHAT IT ENCODES

'R' = renal (usually MILD-to-MODERATE in TTP) and 'N' = neurologic — fluctuating, often dramatic findings: headache, confusion, focal deficits, seizures, or coma from CNS microthrombi. In TTP neuro tends to dominate while renal is comparatively mild.

BOARD TRAP

The organ-emphasis discriminator: SEVERE renal failure with only mild/absent neuro signs (especially a child after bloody diarrhea) favors HUS, whereas prominent NEURO findings with milder renal involvement favor TTP. Choosing HUS for a neuro-dominant adult or TTP for a renal-dominant post-diarrheal child is the planted swap.

8. Pentad rare

WHAT YOU SEE

A green '<25% HAVE FULL PENTAD' callout — only a quarter ever show all five FAT RN features.

WHAT IT ENCODES

Fewer than ~25% of TTP patients ever show the FULL pentad. The modern, exam-correct rule is that unexplained MAHA + thrombocytopenia (a dyad) is sufficient to presume TTP and START plasma exchange — you do not wait for the other three elements to appear.

BOARD TRAP

Waiting for the complete pentad before initiating therapy is the cardinal management error in TTP. Because untreated TTP mortality approaches ~90% and treated is ~10–20%, the right answer is empiric PLEX on the MAHA + thrombocytopenia dyad. Holding treatment 'until more criteria are met' is wrong and lethal.

9. TTP/HUS vs DIC coags

WHAT YOU SEE

A balance SCALE weighing 'NORMAL PT / aPTT / FIBRINOGEN = TTP/HUS NOT DIC' against the DIC side — normal coags tip it to TTP/HUS.

WHAT IT ENCODES

Key lab fork: in TTP/HUS the PT, aPTT, and fibrinogen are NORMAL (and D-dimer normal/mildly up) because the cascade is not the driver. In DIC the PT and aPTT are PROLONGED, fibrinogen is LOW, and D-dimer is markedly elevated. So normal coags + MAHA + low platelets = TTP/HUS; deranged coags = DIC.

BOARD TRAP

The wrong answer treats abnormal coagulation studies as compatible with TTP. Prolonged PT/aPTT, low fibrinogen, and very high D-dimer in a thrombocytopenic patient with schistocytes mean DIC (look for sepsis, malignancy, obstetric catastrophe) — and DIC management (treat the cause, replace factors) differs entirely from PLEX for TTP.

10. Microthrombi

WHAT YOU SEE

The factory CONVEYOR BELT piled with red platelet MICROTHROMBI — the vWF-rich clots churned out across small vessels.

WHAT IT ENCODES

The core lesion is widespread PLATELET-RICH (vWF-rich, fibrin-poor) microthrombi in the small vessels of brain, kidney, heart, and other organs. These both consume platelets (thrombocytopenia) and obstruct/shear flowing RBCs (MAHA), explaining the whole syndrome from one mechanism.

BOARD TRAP

Confusing TTP microthrombi with the fibrin-rich, cascade-driven clots of DIC is the trap. TTP clots are platelet/vWF-rich with a NORMAL fibrinogen, so anticoagulation and factor replacement are not the treatment — PLEX is. Treating TTP as if it were DIC (heparin, FFP for factor repletion only) misses the real fix.

11. Schistocytes = MAHA

WHAT YOU SEE

Fragmented, helmet-shaped red cells captioned 'SCHISTOCYTES = MAHA' — the sheared RBCs that signal microangiopathy.

WHAT IT ENCODES

SCHISTOCYTES (helmet cells, fragmented triangular RBCs) on the peripheral smear are the visual signature of mechanical RBC shearing through microthrombi = microangiopathic hemolytic anemia. Their presence with thrombocytopenia is the trigger to think TTP/HUS/DIC.

BOARD TRAP

Schistocytes are NOT a feature of ITP — if a thrombocytopenia stem shows fragmented red cells, you are out of ITP and into a microangiopathy (TTP/HUS/DIC). The trap is to keep an ITP label once schistocytes appear; conversely, their absence with isolated low platelets and large platelets points back toward ITP.

12. Don't wait — start PLEX

WHAT YOU SEE

A green 'START' button on the line captioned 'MAHA + TCP ENOUGH TO START TREATMENT / DO NOT WAIT FOR PENTAD' — the dyad alone fires the start of plasma exchange.

WHAT IT ENCODES

Action rule: the moment you have unexplained MAHA + thrombocytopenia, START PLASMA EXCHANGE EMPIRICALLY — do not wait for ADAMTS13 results, neuro signs, or the full pentad. Send ADAMTS13 BEFORE the first exchange (so the result is interpretable) but never let the draw delay therapy. Add steroids; consider caplacizumab/rituximab.

BOARD TRAP

Delaying PLEX to wait for the ADAMTS13 activity level to return is the classic fatal mistake — the assay takes days while untreated TTP kills. The correct sequence: draw ADAMTS13 first, then immediately begin empiric plasma exchange. 'Wait for confirmatory testing' is always the wrong TTP answer.

13. Haptoglobin / LDH / bili

WHAT YOU SEE

A hemolysis lab row: HAPTOGLOBIN low (consumed), LDH high (elevated), INDIRECT BILIRUBIN high — the shattered-RBC chemistry panel.

WHAT IT ENCODES

Hemolysis lab panel in TTP: haptoglobin LOW (consumed binding free hemoglobin), LDH HIGH (released from lysed RBCs and ischemic tissue — also tracks disease activity/response), and INDIRECT bilirubin HIGH (heme breakdown), with reticulocytosis. LDH is a useful marker to follow treatment response.

BOARD TRAP

Don't dismiss TTP because one hemolysis marker looks unimpressive — the pattern (low haptoglobin + high LDH + high indirect bili + schistocytes) is what matters, and a markedly high LDH may reflect both hemolysis and tissue ischemia. Attributing the high indirect bilirubin to liver disease and overlooking hemolysis is the distractor.

14. Coombs negative

WHAT YOU SEE

A COOMBS TEST panel marked NEGATIVE 'RULES OUT AIHA' — negative Coombs because the hemolysis is mechanical, not antibody-driven.

WHAT IT ENCODES

The direct Coombs (DAT) is NEGATIVE in TTP because the hemolysis is mechanical (shear), not antibody-mediated. A negative Coombs in a hemolytic, thrombocytopenic patient with schistocytes supports a microangiopathy (TTP/HUS) and excludes autoimmune hemolytic anemia.

BOARD TRAP

A POSITIVE Coombs steers you AWAY from TTP toward autoimmune hemolysis — warm AIHA, or Evans syndrome if combined with immune thrombocytopenia (which is treated with steroids/IVIG, not PLEX). The trap is diagnosing TTP despite a positive DAT; in TTP the Coombs must be negative.